PEAS Framework and Task Environment Analysis

System B: Personalized News Feed Recommender (e.g., Facebook)

Task 1 PEAS Specification

The chosen system is a Personalized News Feed Recommender, modelled on platforms such as Facebook. This agent operates entirely in the digital domain: it observes user behavior across billions of sessions, ranks thousands of candidate posts in real time for each user, and selects the subset most likely to maximize the platform's objectives. Unlike a simple search engine that responds to an explicit query, this agent proactively curates content in the absence of any stated user intent making it one of the most consequential deployed AI systems in the world.

1.1 Performance Measure

Four specific, measurable metrics define this agent's success. First, session engagement rate the proportion of recommended posts that receive a meaningful interaction (click, share, comment, or watch beyond five seconds) is the primary optimization target, typically benchmarked against a baseline of 15-25% depending on content type. Second, feed diversity index measures the breadth of topics, sources, and political perspectives surfaced across a session; a score below a set threshold triggers corrective re-ranking to prevent filter bubble formation. Third, misinformation exposure rate tracks the share of sessions in which the agent surfaces content later fact-checked as false; the internal target is to reduce this rate by at least 30% relative to an unfiltered chronological feed. Fourth, user retention delta measures the change in daily active usage attributable to the feed quality; a sustained engagement gain of even 0.1% at Facebook's scale equates to millions of additional daily sessions. These four metrics are in perpetual tension with each other, which is precisely what makes this system challenging to optimise.

1.2 Environment

The agent operates within a layered digital environment with no physical analogue. At the user level, it contends with diverse demographics, languages, cultural contexts, and rapidly shifting moods a piece of political satire may delight one user and deeply offend another with an identical engagement history. At the platform level, the environment includes the full corpus of user-generated content (posts, images, videos, links), advertiser bidding systems competing for feed slots, and regulatory constraints that vary by jurisdiction content legal in the United States may be prohibited in Germany or India. At the network level, the agent must operate with near-zero latency across data centres on multiple continents, serving personalised feeds to hundreds of millions of concurrent sessions without perceptible delay. Trending events elections, disasters, celebrity news create sudden, non-stationary spikes in content volume that stress the ranking pipeline in real time.

1.3 Actuators

The agent's actuators are entirely digital but have real-world consequences. Its primary actuator is the feed ranking and display engine, which reorders and selects from a candidate pool of thousands of posts to determine precisely which content appears, in what order, and in what format (full text, image preview, video autoplay, or link card). A notification dispatch system acts as a secondary actuator, pushing alerts about trending content or friend activity to re-engage users who have left the platform. The content demotion and promotion mechanism suppresses or amplifies individual posts by adjusting their distribution score effectively making a post invisible to most users or virally spreading it to non-followers. Finally, an A/B testing framework continuously deploys minor ranking model variants across user segments, using the resulting engagement data to update model weights automatically without human intervention.

1.4 Sensors

The sensor suite is behavioural and contextual rather than physical. Explicit interaction logs likes, shares, comments, and saves provide direct supervisory signal but represent only a small fraction of true user preferences. Implicit engagement signals are far richer: scroll velocity (slowing over a post signals interest even without a click), dwell time (the number of seconds a post remained on screen), video watch percentage, and hover events on images. User profile and network data age, location, language, friend connections, group memberships, and historical topic affinities provide personalisation context. Content-side sensors include natural language processing models that extract topic, sentiment, and factual credibility scores from post text, and computer vision models that classify image and video content. External signals such as real-world trending topics from news APIs and regional event calendars are ingested to contextualise session-level recommendations.

Task 2 Environment Classification

The following six-dimensional classification is grounded directly in how a news feed recommender operates in production not in abstract definitions. Each property carries non-trivial engineering implications for platform designers.

Dimension	Classification	One-Sentence Justification
Observable	**Partially Observable**	The agent cannot directly observe why a user clicked it only sees the click event itself, not the underlying intent, mood, or whether the content was actually consumed rather than just opened.
Agent Type	**Multi-Agent**	Millions of users interact simultaneously, their collective engagement data feeds back into model retraining, and advertiser bidding systems compete in the same environment as the recommender.
Outcome	**Stochastic**	Even a post highly similar to previously liked content may be ignored on a stressful Monday morning user receptiveness is mood-dependent and not deterministically predictable from historical data.
Time Horizon	**Sequential**	Each recommendation shapes future user preferences and the content ecosystem; pushing emotionally charged posts to boost short-term engagement may radically alter the user's long-term content diet.
World Change	**Dynamic**	Trending topics, breaking news, and user emotional states evolve continuously while the agent is ranking posts the world is never frozen between decisions.
State Space	**Continuous**	Engagement probabilities, sentiment scores, attention-time estimates, and relevance weights are all real-valued signals, not discrete categories.

Task 3 Critical Analysis

3.1 Most Difficult Engineering Challenge

Of the six environment properties, the sequential nature of the environment is the most difficult challenge for engineers building this system. The difficulty is not technical in the traditional sense   it is causal and temporal. Every recommendation made today influences user preferences tomorrow. A feed algorithm optimised for short-term clicks gradually narrows a user's information diet: political content that generates outrage receives more clicks, so the algorithm surfaces more of it, which reinforces the user's engagement with outrage-inducing content, which further skews their interest profile, and so on. This feedback loop   what researchers call a preference amplification spiral   is invisible in any single session's metrics but devastating across months of interaction.

The engineering challenge is that the standard toolkit for sequential decision-making (reinforcement learning, multi-step planning) requires either a ground-truth reward signal delayed by weeks or months, or a reliable simulation of how user preferences evolve   neither of which currently exists at production scale. Platforms can measure click-through rates in milliseconds, but measuring whether a user's long-term psychological wellbeing improved or declined as a result of their feed is an open research problem.This temporal credit assignment gap   knowing which past recommendation decisions caused today's user behaviour   is arguably the defining unsolved problem in recommendation system engineering.

3.2 Utility Function and Trade-off Analysis

A realistic utility function for this system can be written as:

$$U = \alpha \cdot \text{Engagement\_Score} - \beta \cdot \text{Misinformation\_Risk} - \gamma \cdot \text{Outrage\_Amplification}$$

Here, Engagement_Score captures the weighted sum of user interactions (clicks, shares, and comments carry different weights reflecting their depth of engagement). Misinformation_Risk is a real-time credibility score generated by the platform's fact-checking pipeline   posts from repeatedly flagged sources receive a higher penalty coefficient. Outrage_Amplification penalises content whose emotional valence data predicts it will elicit strong negative affect, measured via sentiment analysis and historical sharing patterns of similar posts. The weights α, β, and γ define the platform's implicit editorial values.

If α is doubled to, say, 2.0 while β and γ remain at their original values, the ranking model becomes disproportionately rewarding to engagement at any cost. In practical terms, posts that generate outrage, fear, or tribal identity conflict score highest on engagement metrics while frequently carrying misinformation. The algorithm would begin systematically promoting these posts   not because any engineer instructed it to, but because the inflated α makes high-engagement-misinformation posts score above low-engagement-accurate posts in the utility calculation. Within weeks, the feed would exhibit measurably higher rates of divisive and false content, and advertiser brand-safety complaints would increase. This is not a hypothetical   internal studies at major platforms have documented exactly this dynamic, which is why ethical AI governance frameworks now mandate regular audits of recommendation weight configurations rather than treating them as purely technical parameters.

Reference: Russell, S., & Norvig, P. (2022). Artificial Intelligence: A Modern Approach (4th ed.). Pearson

Task 4: Structured JSON Representation


```
{
  "system_name": "Personalized News Feed Recommender",
  "peas": {
    "performance_measures": [
      "Session Engagement Rate",
      "Feed Diversity Index",
      "Misinformation Exposure Rate",
      "User Retention Delta"
    ],
    "environment": "Large-scale social media ecosystem containing users, content creators, advertisers, regulations, and rapidly changing real-world events.",
    "actuators": [
      "Feed Ranking and Display Engine",
      "Notification Dispatch System",
      "Content Promotion and Demotion Mechanism",
      "A/B Testing Framework"
    ],
    "sensors": [
      "Explicit Interaction Logs",
      "Implicit Engagement Signals",
      "User Profile and Network Data",
      "Content Analysis Models",
      "Computer Vision Models",
      "External Trending Signals"
    ]
  },
  "environment_classification": {
    "observable": {
      "choice": "Partially Observable",
      "justification": "The agent cannot directly observe user intent and relies on indirect behavioral signals."
    },
    "agent_type": {
      "choice": "Multi-agent",
      "justification": "Users, advertisers, and platform systems interact simultaneously in the environment."
    },
    "outcome": {
      "choice": "Stochastic",
      "justification": "User behavior is influenced by unpredictable contextual factors."
    },
    "time_horizon": {
      "choice": "Sequential",
      "justification": "Current recommendations affect future user preferences and actions."
    },
    "world_change": {
      "choice": "Dynamic",
      "justification": "News, trends, and user interests change continuously."
    },
    "state_space": {
      "choice": "Continuous",
```

"justification": "Relevance scores and engagement estimates are continuous values."
}
},
"utility_function": "U = α × Engagement_Score − β × Misinformation_Risk − γ × Outrage_Amplification"